

Plant-induced changes in the rhizosphere of maize and wheat

II. Complexation of cobalt, zinc and manganese in the rhizosphere of maize and wheat*

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Summary Water-soluble ^{14}C -labelled organic material, released into soil from roots of wheat and maize plants, was recovered with a mild percolation technique, without disturbing the root-soil interface. Extraction yields were relatively high for the ^{14}C materials (up to 11% of residual soil ^{14}C for 6 weeks maize) illustrating the water soluble character of the freshly added material. The complexation potential of the soil extracts was evaluated by adding ^{57}Co , ^{65}Zn and ^{54}Mn to the extract and determining their distribution among the organic fractions by a gel filtration technique. The results show that within four weeks a micro-environment is created around a plant root, characterized by an accumulation of root-derived organic materials. In parallel with this time dependent accumulation, a gradual shift from ionic metal to higher molecular weight forms occurred. The three metals were increasingly complexed throughout the growth period. Extracts from a fallow soil complexed minor amounts of the added tracer (6.4%; 1.9% and 0.2% for ^{57}Co , ^{65}Zn and ^{54}Mn respectively) while cropped soil extracts after 6 weeks complexed 61%; 16% and 6% of respectively Co, Zn and Mn in the case of maize, and 31%, 15% and 1% in the case of wheat. Although the effects are most pronounced in the case of Co and maize, evidence is given for similar effects with wheat and the other metals. The results indicate that the physicochemical status of transition metals in the rhizosphere is entirely different from that in the bulk soil.

Introduction

It is well established that an adequate supply of nutrients to plant roots is not merely a function of the total amount of the particular nutrient in the soil, but also — and perhaps even more — of the rate of

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replenishment of the soil solution by the nutrient pool. Replenishment depends on the physical, chemical and biological characteristics of a soil and is different for each element. The discrepancy between total and available levels of a nutrient is most striking for transition metals, many of them being important trace elements for plant nutrition. This explains the numerous attempts to develop chemical extractants which would extract the plant-available fraction in a representative way *i.e.* some sort of mimicry of what is believed to happen in nature. Initially, deficiency problems motivated these investigations, but the pollution of soils by heavy metals and radionuclides gave a new impulse to this research.

The classical and essentially empirical approach is to correlate extracted amounts with the content present in plants grown on the same soil. More meaningful extraction schemes are based on the idea that the uptake of ions from the soil solution might be controlled or at least mediated by the plant root itself by releasing specific components that bind these ions through chelation or complexation reactions⁹. Micro-nutrient deficiencies have been effectively remedied by soil amendments of various synthetic chelates¹³. Chelation enhances the mobility or solubility of the deficient element, either indigenous or added as fertilizer, leading to an improved availability to the plant root.⁴.

Simulating root action, synthetic chelates proved to be successful in availability assessments, in the belief that the extracted amounts are derived from the same pool as the one explored by roots.

The *in situ* speciation, or the distribution of a metal over the different molecular species in the soil solution, remains unknown by this essentially pragmatic approach. Metals in soils occur as simple ions, hydrolyzed species¹⁷, chelated ions¹, precipitates⁶, adsorbed ions, possibly in different oxidation states.

Unravelling these equilibria necessitates a compromise between two contrasting points of interest. On the one hand, one aims to determine the very "in situ" speciation free from artefacts and on the other hand, one is forced to disturb the system for sampling, which cannot be done without altering the original soil solution composition. Perhaps the most reliable way of sampling soil solution is the immiscible displacement-centrifugation technique¹¹. Although the soil solution thus collected is quite representative and therefore of great value in plant nutrition studies, the method is not entirely troublefree when plant root induced changes are to be examined. The most important reason for this is that the root affected amount of soil is only a very small part of the "bulk soil" and is in intimate contact with the root surface. Attempts to isolate rhizosphere soil mechanically from roots and "non-rhizosphere" soil are always subject to criticism.

In the present paper, the effect of growing plant roots on the speciation of Co, Zn and Mn is investigated by a non-conventional approach. Growing plants in a $^{14}\text{CO}_2$ atmosphere resulted in ^{14}C -labelling of the root-derived soil organic matter. The water-soluble components of ^{14}C -labelled material were recovered by leaching soil columns containing undisturbed roots, with distilled water. The soil extracts were then subject to a characterization and evaluation of their complexing capacity, assuming that similar reactions occur in the rhizosphere. Generally the molecular weight of a complexed metal species is higher than that of the simple ionic species. This size difference can be used to separate both species. Size grouping of biological molecules can be performed by either ultrafiltration or by gel filtration chromatography (GFC). The major advantage of these two methods based on size differences is that separations can be performed under realistic conditions. In this work preference was given to gel filtration being less time consuming than ultrafiltration.

The use of gel filtration in speciation studies is not new and particularly the association of transition metals with humic acids or fulvic acids in aquatic or soil ecosystems has repeatedly been demonstrated by GFC^{2, 3, 5, 8, 14, 15, 18}. It must be emphasized, however, that GFC when applied to humic type components, as prevailing in soil extracts, results in a fractionation based on both molecular sieving and adsorption phenomena. Whenever accurate data on molecular weight are the goal of the gel filtration experiment, the method is therefore not appropriate⁷. In the present set-up, however, this question is rather academic since adsorption phenomena can be of additional value in separating ionic forms from the complexed forms.

When labile complexes are the point of interest, however, the exchange sites on the gel matrix may stimulate the complete dissociation of the complex since the higher molecular weight species in equilibrium with the ionic metal are immediately removed by the chromatographic process. The behaviour of inorganic solutes on gel filtration columns can in fact not be described in terms of steric exclusion only. Many important side effects like ion exclusion, adsorption, counter anion and co-ion influence are reviewed by several authors^{12, 16, 19}. Relevant to this study is that the appearance of a particular metal at a different elution volume suggests different molecular species of this metal. Gel filtration experiments of soil extracts labelled with a metallic tracer provide fingerprints of the speciation pattern relevant for the soil solution, sensitive enough to detect plant root induced changes.

Materials and methods

Growth conditions and harvest procedures have been described in Part I¹⁰. At both harvests, after 4 and 6 weeks of plant growth, six soil columns with plants were removed from the growth chamber. Three of them were used to collect root-free soil in order to assess the total amounts of root-derived material. The three other columns were saturated with distilled water with a peristaltic pump (250 cm³/h). Upon saturation, percolation was initiated by siphon draining and continued until a 600-cm³ leachate was collected. The same procedure was performed for a fallow-column. Extracts were divided in 100 cm³ batches and stored at -18°C. Shoot and root material were then removed from the soil columns and processed as before¹⁰.

Before use, any colloidal material, precipitates and bacteria were removed from the extract by membrane filtration (pore diameter 0.45×10^{-6} m). The extracts were concentrated five times by freeze-drying and assayed for total carbon (T.O.C. analyser, Beckman 915A) and ¹⁴C (LSC, Nuclear Chicago, Mark I). Samples for metal complexation studies, were prepared from these concentrated extracts by adding a known amount of carrier-free radionuclides dissolved in 300 µl of a 3.1×10^{-3} M NaN₃ solution, to 900 µl of the extract. The radioactivity of the additions was 6109 Bq of ⁶⁵Zn, 3139 Bq of ⁵⁴Mn and 733 Bq of ⁵⁷Co (in one case 317 Bq of ⁵⁷Co, 4 weeks maize). Samples were equilibrated for at least 48 hours.

Fractionation

Standard chromatography equipment was used throughout, consisting of a peristaltic pump (LKB 10200 PERPEX), UV monitor (LKB 8300 UVICORD II), recorder and fraction collector (LKB-7000 ULTRORAC). Sephadex G-15 (Pharmacia Fine Chemicals) was used in columns of 80 cm length and 0.79 cm² cross section. Elution during fractionation was performed with a 3.1×10^{-3} M NaN₃ solution, while the non-complexed species were eluted afterwards with a 10^{-1} M CaCl₂ solution, the flow rate being 12 cm³/h for both eluents. Regeneration of the gel was achieved by changing back to 3.1×10^{-3} M NaN₃ until the effluent of the column was Cl⁻ free (acidified AgNO₃ solution test). Each soil extract was passed on the gel at least four times. In a first run, without metallic tracer, the C/¹⁴C fractionation pattern was determined, while in the three other runs the distribution of the added radionuclides among the organic fractions was determined. In each case samples of 1 cm³ were injected into the column and 1 cm³ fractions were collected for ¹⁴C, C or γ-activity determination (respectively by LSC (Nuclear Chicago Mark I), T.O.C. analyser (Beckman 915A) and γ-spectrometer (Philips PW 4003)). With CaCl₂ as the eluent, one batch was collected to determine the γ-activity of the non-complexed species. Only when recovery of the added isotope was complete, a next sample was introduced into the column.

Results

Plant growth

The dry weights, ¹⁴C contents and specific activities of the maize and wheat plants grown in the ¹⁴CO₂ environment have been given in Part I¹⁰. Relevant to this experiment are the relative amounts of soil ¹⁴C as the percentage of the total amount recovered in shoot, root and soil material at harvest. For 4- and 6-week old maize plants these values were 3.5 and 1.5% respectively and for wheat 2.4 and 2.0% respectively.

Extraction

The results of the percolation procedure are given in Table 1. The pH of the cropped column extracts varied in the range 6.8 to 7.1, compared

Table 1. Mineral composition (Ca, Mg, Zn, Fe, Mn and Co concentration); carbon and ^{14}C content of the root-soil extracts obtained by siphon-draining. Total volume of leachate was 600 cm^3

	Root-soil extract with				
	Fallow	Maize after		Wheat after	
		4 weeks	6 weeks	4 weeks	6 weeks
pH	5.7	6.9	6.8	6.8	7.1
C $\mu\text{g}/\text{cm}^3$	98.6	107.6	136.8	122.2	123.7
^{14}C Bq/ cm^3	0.12	18.3	172.5	28.9	39.4
Specific activity, kBq $^{14}\text{C}/\text{mg C}$	0.001	0.17	1.26	0.24	0.32
Ca $\mu\text{g}/\text{cm}^3$	282	ND	130	115	62
Mg $\mu\text{g}/\text{cm}^3$	30	ND	21	21	18
Zn $\mu\text{g}/\text{cm}^3$	1.19	0.69	0.26	0.13	0.12
Fe $\mu\text{g}/\text{cm}^3$	0.42	1.54	1.54	1.03	0.83
Mn $\mu\text{g}/\text{cm}^3$	0.84	2.35	0.52	0.17	0.11
Co ng/ cm^3	29	ND	38	19	16

to 5.7 for the fallow column extract. The concentration of total organic carbon ranged between 98.6 and $136.7\text{ }\mu\text{g C}/\text{cm}^3$. The total carbon content of the soil was 1.3–1.5%, the total amount present in each column was thus 19.5–22.5 g C. The extracts contained 59.2–82.1 mg C or 0.3–0.4% of the total soil C. As expected, the amounts of ^{14}C in the extract increased with plant production in a roughly proportional way.

Of the ^{14}C present in shoots, roots and soil, 1.5% was present in soil when maize was grown for 6 weeks. This is a total quantity of $0.97 \pm 0.05\text{ MBq}$ per column. ^{14}C content of the soil extract was $172.5\text{ Bq}/\text{cm}^3$ or 0.103 MBq per column. The extraction efficiency was thus $0.103/0.97 \times 100 = 10.7\%$. From the respective activities of the extracts and the ambient specific activity in the growth chamber ($4.5\text{ kBq }^{14}\text{C}/\text{mg C}$) one can calculate that 28% of the extracted carbon is root-derived in the case of the 6 weeks maize column extract. In general, the concentrations of metals were decreasing somewhat with plant growth, except for Fe, the concentration of which increased with plant growth, and for Co and Mn, whose concentrations varied somewhat unpredictably.

Fractionation

Generally, the recovery of ^{14}C compounds from the gel bed within the theoretical total volume of the Sephadex gel column exceeded 80%. (Between K_{av} 0–1). The remaining fraction was eluted later, indicating some interaction with the gel. The combination of both eluents, $3.1 \times 10^{-3}\text{ M NaN}_3$ and 10^{-1} M CaCl_2 also enabled a complete recovery of the added metallic radionuclides. Since ionic forms of the metals under study did not elute within the theoretical total volume under the

Table 2. Amounts complexed of the radionuclides ^{57}Co , ^{65}Zn and ^{54}Mn , added to different soil extracts. The amount complexed Q_c is given in % according to $Q_c = A_o \times 100/A_o + A_i$

Sample origin	^{57}Co	^{54}Mn	^{65}Zn
Fallow	6.4	0.2	1.9
4 weeks maize	58.6	3.5	11.8
6 weeks maize	60.6	5.9	15.8
4 weeks wheat	22.7	0.4	3.7
6 weeks wheat	30.5	0.8	15.2

applied conditions, the presence of metal activity within the organic fractions indicated the presence of organometallic associations. The amounts complexed are then readily obtained from the following formula:

$$Q_c = \frac{A_o \times 100}{A_o + A_i}$$

where Q_c is the amount complexed in %, A_o is the γ -radioactivity in the organic fractions eluted between K_{av} 0–1 and $A_o + A_i$ equals the amount added to the column, A_i being the radioactivity of the non-complexed species eluted with CaCl_2 $10^{-1} M$ (Table 2).

For all radionuclides and all treatments, the amounts complexed were higher in the cropped soil column extracts than in those from the fallow columns. This effect was most striking for the combination ^{57}Co -maize where 58.6 and 60.6% of the added amounts were complexed after four and six weeks plant growth respectively, compared with 6.4% in the fallow column extract. In all cases a higher complexation percentage was measured with maize than with wheat. Manganese seemed to be the least affected by plant growth. The 0.2% complexed in the fallow column increased to maxima of 0.8% and 5.9% for 6 weeks wheat and maize respectively.

The chromatograms of the fallowed and maize columns soil extracts with or without ^{57}Co will be discussed in more detail. The percolate of a fallow column is considered first. The $\text{C}/^{14}\text{C}$ distribution among 1 cm^3 fractions is given in Fig. 1. No detectable amounts of ^{14}C were present in the different fractions, since only traces were present in the initial extract (0.1 Bq/cm^3). Total organic carbon concentrations corresponded well with the 254 nm absorbance pattern, except for the last peak ($K_{av} = 1$).

Fig. 2 shows the chromatogram of the fallow extract, spiked with ^{57}Co . As with the other additions, only minor amounts of metal were co-eluted with the native soil organic matter components (Table 2).

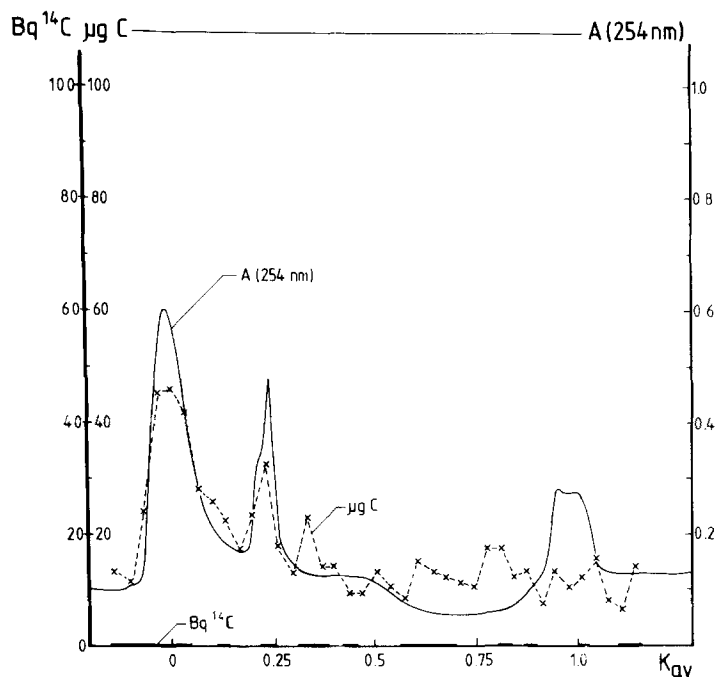


Fig. 1. Distribution of carbon and ^{14}C and absorbance (254 nm) obtained by gel filtration of a fallow column extract (H_2O) on Sephadex G-15. Column length 80 cm; cross section 0.79 cm^2 ; flowrate $12\text{ cm}^3/\text{h}$; sample volume 1.0 cm^3 ; eluent $3.1 \times 10^{-3}\text{ M NaN}_3$.

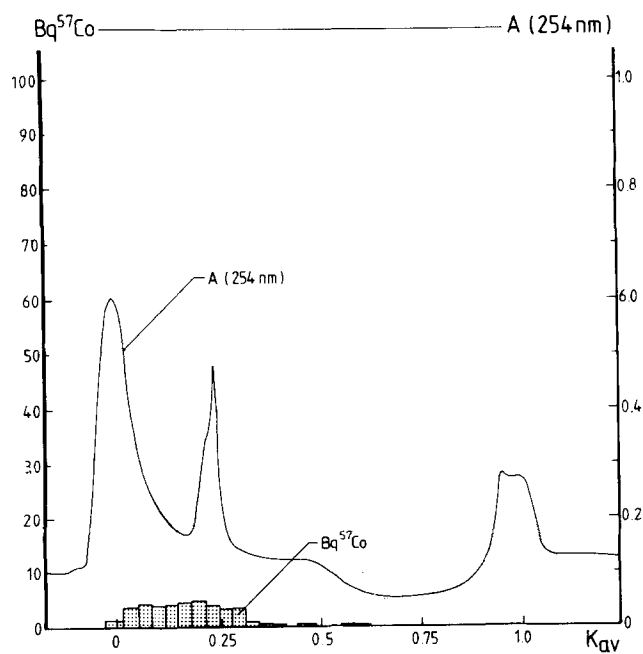


Fig. 2. Distribution of ^{57}Co and absorbance (254 nm) obtained by gel filtration of a fallow column extract (H_2O) on Sephadex G-15. Column length 80 cm; cross section 0.79 cm^2 ; flow rate $12\text{ cm}^3/\text{h}$; sample volume 1.0 cm^3 ; eluent $3.1 \times 10^{-3}\text{ M NaN}_3$.

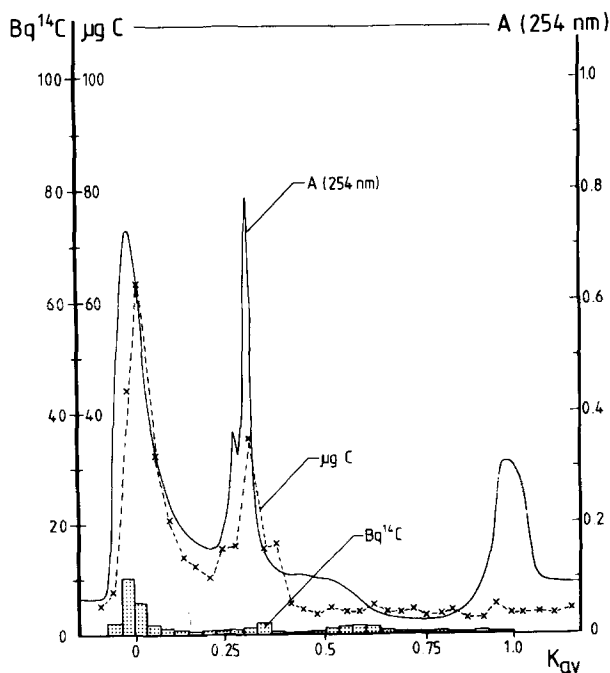


Fig. 3. Distribution of carbon and ^{14}C and absorbance (254 nm) obtained by gel filtration of a 4 weeks maize root-soil column extract (H_2O) on Sephadex G-15. Column length 80 cm; cross section 0.79 cm^2 ; flow rate $12\text{ cm}^3/\text{h}$; sample volume 1.0 cm^3 ; eluent $3.1 \times 10^{-3}\text{ M NaN}_3$.

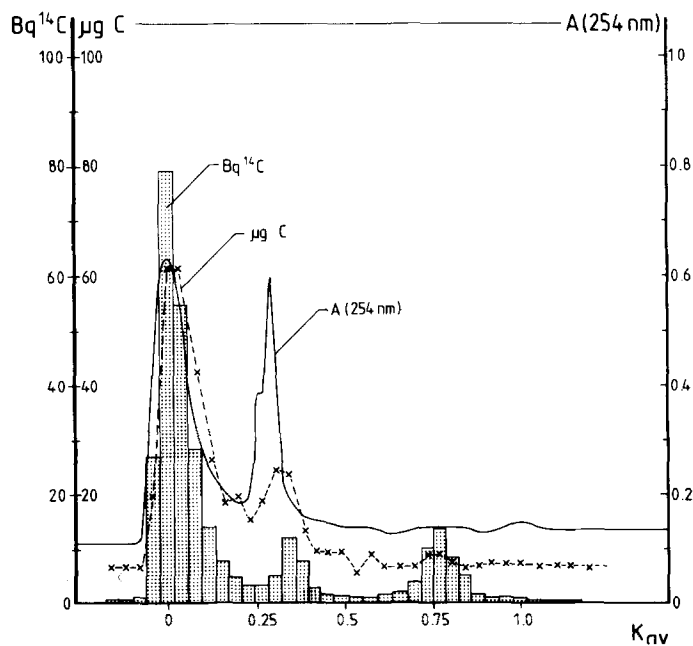


Fig. 4. Distribution of carbon and ^{14}C and absorbance (254 nm) obtained by gel filtration of a 6 weeks maize root-soil column extract (H_2O) on Sephadex G-15. Column length 80 cm; cross section 0.79 cm^2 ; flow rate $12\text{ cm}^3/\text{h}$; sample volume 1.0 cm^3 ; eluent $3.1 \times 10^{-3}\text{ M NaN}_3$.

Results from the maize experiment are given in Figures 3, 4, 5 and 6. The $C/^{14}C$ distribution at both harvests (Fig. 3 and 4 respectively) shows that in both cases the main part of the ^{14}C was eluted immediately behind the void volume, indicating a molecular weight exceeding 1500. Absorbance maxima at $K_{av} = 0$ and 0.3 at four weeks and 6 weeks again agreed well with the total organic carbon concentrations. Generally, the ^{14}C distribution traced this pattern with the exception of a peak with $K_{av} = 0.6$ in the 4 weeks percolate and at $K_{av} = 0.8$ in the 6 weeks percolate. In the 6 weeks percolate the absorbance peak at $K_{av} = 1$ disappeared. The near absence of carbon in these fractions and the elution at $K_{av} = 1$ suggests the presence of inorganic salts, as was confirmed by NO_3^- analysis.

The distribution of ^{57}Co among the several fractions of the 4 weeks and 6 weeks percolate is given in Figures 5 and 6 respectively. The increase in amounts complexed after 4 and 6 weeks plant growth was accompanied by a qualitative change in the speciation pattern. After 4 weeks, the ^{57}Co distribution was not very pronounced as its maximum activity agreed neither with a maximum in total carbon, nor with a

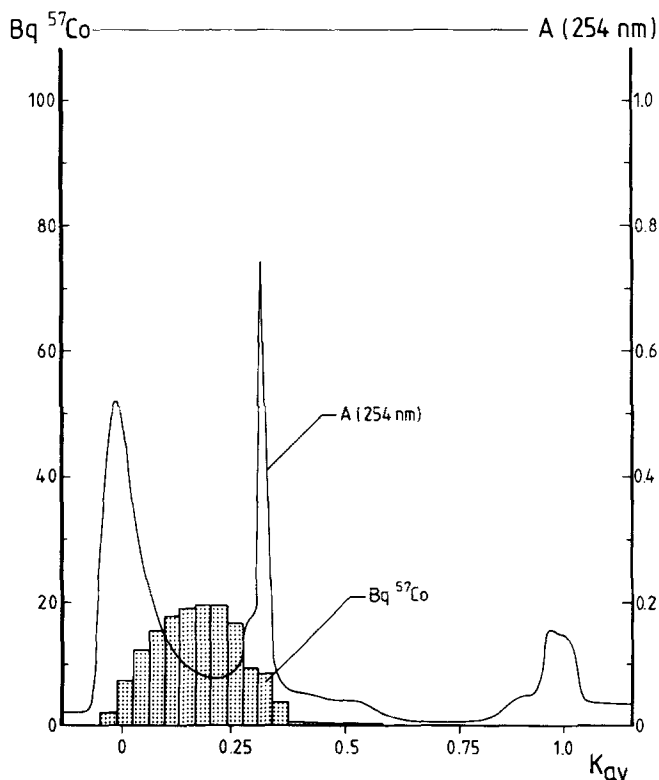


Fig. 5. Distribution of ^{57}Co and absorbance (254 nm) obtained by gel filtration of a 4 weeks maize root-soil column extract (H_2O) on Sephadex G-15. Column length 80 cm; cross section $0.79\ cm^2$, flow rate $12\ cm^3/h$; sample volume $1.0\ cm^3$; eluent $3.1 \times 10^{-3}\ M\ NaNO_3$.

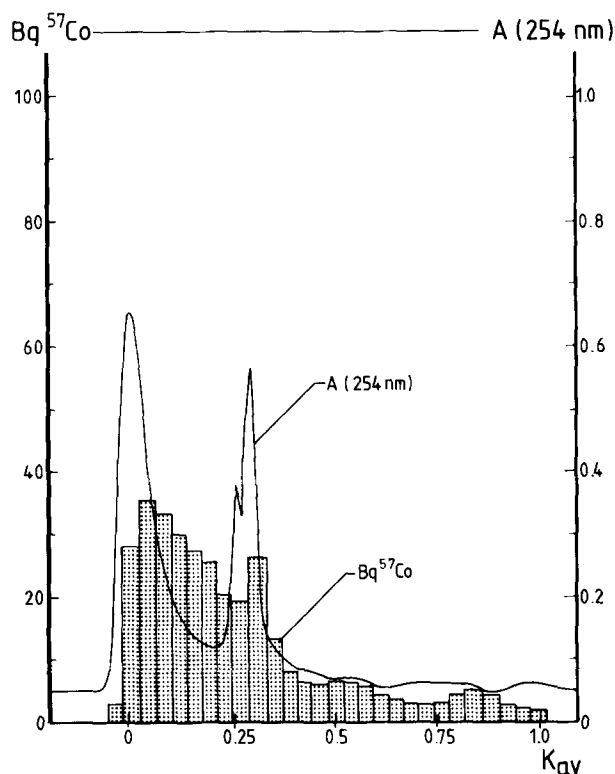


Fig. 6. Distribution of ^{57}Co and absorbance (254 nm) obtained by gel filtration of a 6 weeks maize root-soil column extract (H_2O) on Sephadex G-15. Column length 80 cm; cross section 0.79 cm^2 , flow rate $12\text{ cm}^3/\text{h}$; sample volume 1.0 cm^3 ; eluent $3.1 \times 10^{-3}\text{ M NaN}_3$.

maximum in ^{14}C -activity. At 6 weeks a much more specific association was measured. The ^{57}Co activity showed definite maxima at K_{av} values of 0.05; 0.30; 0.50; and 0.85. The first two peaks were quantitatively the most important and agreed well with the total carbon and ^{14}C maxima with K_{av} values of 0 and 0.3 respectively. For ^{65}Zn and ^{54}Mn similar trends were recorded, although the amounts involved in relative terms were smaller than with ^{57}Co .

Increases occurred in the amounts complexed when going from fallow to cropped soil column extracts as shown in Table 2.

Discussion

The percolation technique with water has provided a mild extraction procedure to isolate the ^{14}C -labelled organic matter derived from plant roots, and it should have avoided artefacts involved in conventional root-soil separation techniques. The rather high soil/water ratio during

extraction is another advantage, diminishing the risk of peptisation of soil colloids or plant cell leakage by osmosis. With water as extractant, no foreign elements were introduced into the system and the solutes extracted were in fact present in a soluble form in the soil solution.

As could be expected, the ^{14}C -activity in the soil column percolates runned parallel with plant growth. The root-derived materials were extracted 26 times more efficiently than the competing native soil organic components as evidenced by ^{14}C -content. The difference in extraction yield between native soil components and root-derived material is an attractive feature of the extraction technique. Taking into account that only a small fraction of the soil is in contact with the root, it can be postulated from these data that the root-originating materials must prevail in the root-soil contact zone, even in cases where a high concentration of native soil components can be expected. The large difference in extractability of C and ^{14}C is not surprising, since fresh plant derived materials, to a great extent "simple" molecules, are more soluble in water than the native, more recalcitrant organic matter. Undoubtedly, the kind of soil used, having a low clay mineral content is a key factor in the good performance of this percolation method. However, this does not limit the present findings to this soil, but it is acknowledged that it is easier to work with. Gel filtration of these soil percolates, labelled with selected radionuclides, shows that root-derived products successfully compete with native soil organic matter for complexation reactions. Evidence for this is given by the fact that the total amounts of metal complexed as assessed by a gel filtration method are clearly increasing with plant growth. This effect is most pronounced for ^{57}Co in maize where, after 6 weeks growth, 60.6% of the added amount is associated with the root-derived components, as compared to 6.4% which is associated with native soil organic components in a fallow soil extract. However, some care in the handling of these figures is necessary since equilibria may shift during the gel filtration process. Presumably such artefacts, due to interaction of sample molecules with the gel matrix, will lead to an underestimate of the amounts complexed. It was shown that especially ionic metal species are retarded on the gel bed, by ion exchange mechanisms. Complete dissociation of some labile complexes may be the final result of such interactions, leading to a reduction in the percentages complexed.

The shift from ionic species to higher molecular weight species of the three metals under study occurs in parallel with the time dependent, increased presence of root-derived components. At both harvests, and for all metals the complexed amounts are lower for the wheat experiment, than for the maize experiment. This suggests that different types

of materials are involved. It appears that after 4 weeks, the wheat root-derived material is less competitive with the native soil organics than the maize derived material. In spite of the higher amounts present the wheat root material complexes far less metals than the maize root material (Table 2). The composition of the root-derived material is also time dependent, with varying proportions of complexing agents, each of them with a different turnover rate and a different affinity for each metal. The present methodology demonstrates only the overall effect at one given moment during the growth period, although some qualitative differences can be deduced in the case of ^{57}Co when comparing the speciation patterns at 4 and 6 weeks respectively.

In general, bulk extraction techniques fail to demonstrate rhizosphere effects on the physico-chemical form of plant nutrients. Though the rhizosphere is an unavoidable link in the transfer from soil to plant and, as evidenced, a modification of the chemical form during passage through this interface is likely to occur. The accumulation of root-originating materials in the soil solution, and undoubtedly to a larger extent in the soil-root interface, clearly points to the generation of a micro-environment with properties quite different from the bulk soil. Therefore, the change in physicochemical status of the metals under study suggests that their availability and mobility cannot be estimated from bulk soil data.

Generally, an accumulation of chelating agents in the soil results in a facilitated nutrient supply, particularly with respect to trace elements. This is caused by the enhanced solubility and mobility introduced by chelation or complexation reactions. On the other hand, uptake rate of a complexed metal is generally lower than that of an ionic species. Whether the amount of metal finally taken up by plants is affected or not, depends on which mechanism prevails. The present evidence for rhizosphere complexation processes has obvious application to studies concerned with long-term behaviour of transition metals in soil (micro-nutrients, heavy metals, radiocontaminants).

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